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National Institute of Technology, Raipur (Chhattisgarh), India

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Minor Project Final Presentation — November 2026

Spatial-Guided Spectral-Channel Fusion Network for Hyperspectral Remote Sensing Image Analysis

Under the Guidance of

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B.Tech Seventh Semester, Nov 2026

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Outline

- Recap: Problem Statement & Research Objectives
- SGSCF-Net Architecture Overview
- Module Design: SBAM, CBSA, SCFG
- Hybrid Loss Function
- Experimental Setup & Benchmark Datasets
- Results: Indian Pines, Pavia University, Salinas Valley
- Classification Maps Comparison
- Computational Efficiency & Parameter Analysis
- Ablation Study & Depth Sensitivity
- Hyperparameter Grid Search (60 Runs)
- Data Scarcity Stress Testing (1%–10%)
- Cross-Dataset Summary & Key Achievements
- Future Work & Research Directions

Recap: Problem Statement & Research Objectives

Problems Identified (Mid-Semester)

- Hughes Phenomenon: High bands + low samples = overfitting
- Memory Explosion: 3D-CNNs & Transformers exceed 85M+ parameters
- Feature Entanglement: Spatial + spectral mixing blurs class signatures
- Class Imbalance: Minority classes (Oats: 20 samples) collapse to 0%
- Unfiltered Noise: Atmospheric absorption corrupts raw band features

Our Research Objectives

- Build ultra-lightweight architecture (~308K params, 9.1 MFLOPs)
- SBAM: Adaptive band weighting to filter atmospheric noise at input
- CBSA: Complementary gating to prevent feature duplication
- SCFG: Covariance outer products for dynamic channel calibration
- Joint Spread-Focal Loss for minority class protection

What We Delivered (This Presentation)

- Fully implemented & trained SGSCF-Net across 3 benchmark HSI datasets (Indian Pines, Pavia University, Salinas Valley)
- 120+ GPU training runs: 60-run patch/batch grid search + 12 data scarcity stress tests + 24 baseline benchmark runs
- Competitive or superior accuracy vs. 8 SOTA baselines while using only 308K parameters (vs. 6.4M for SSAN, ~610K for ViT)
- Complete 31-page ACM-format research paper with per-class breakdowns, classification maps, ablation studies, convergence analysis

SGSCF-Net: End-to-End Architecture Overview

Stage 1
SBAM
Noise Filtering

Raw HSI patch X
→ Band attention
→ Recalibrated $\tilde{X}$

→

Stage 2
Dual 3D Entry
+ CBSA Gate

1×1×7 spectral branch
3×3×7 spatial branch
α-gating fusion

→

Stage 3
Spectral
ResBlocks ×2

1×1×7 Conv3D
+ ECA-3D attention
+ residual skip

→

Stage 4
SCFG
Covariance Gate

Adaptive 3D pool
$G = f_{ch} \otimes f_{sp}$
Frobenius norm → gate

→

Stage 5
Spatial ResBlocks
+ Classifier

3×3×128 Conv3D
+ GAP → FC head
Spread+Focal Loss

Input: $X \in \mathbb{R}^{(B\times1\times P\times P\times C)}$ → SBAM recalibration → Dual-branch CBSA → Spectral deepening → SCFG covariance gating → Spatial refinement → $\hat{y} \in \mathbb{R}^{(B\times C_cls)}$

Module 1: Spectral Band Attention Module (SBAM)

What SBAM Does

- Operates on RAW spectral bands BEFORE any convolution
- Adaptive 3D average pooling compresses P×P spatial grid
- Squeeze MLP bottleneck (reduction r=8) learns inter-band weights
- Sigmoid gate w ∈ (0,1)^C suppresses noisy atmospheric bands
- Hadamard product: $\tilde{X} = X \odot w$ (channel-wise reweighting)
-
- Key Equations:
- $z_c = (1/P^2) \sum X(b,1,u,v,c)$ [spatial pooling]
- $w = \sigma(W_2 \cdot \delta(W_1 z + b_1) + b_2)$ [MLP bottleneck]
- $\tilde{X} = X \odot E(w)$ [band filtering]

Why It Matters

- Conventional CNNs treat ALL bands equally — noise propagates
- Water vapor absorption at 1.4μm, 1.9μm corrupts features
- SBAM acts as a learnable spectral firewall at network entry
- Parameters: only 10,225 (3.32% of total network)

Design Rationale vs. SENet / CBAM

- SENet/CBAM recalibrate AFTER deep feature extraction
- SBAM recalibrates BEFORE convolution → noise never enters network
- No reinforcement learning or outer optimization loops needed
- End-to-end differentiable with standard backpropagation

Module 2: Complementary Branch Spectral Attention (CBSA)

Dual-Branch Design

- Branch a: 1×1×7 Conv3D → pure spectral inter-band transitions
- Branch b: 3×3×7 Conv3D → joint spatial-spectral context
- Both: K=16 filters, spectral stride=2, BN + ReLU
-
- Complementary Gating Mechanism:
- $U = a + b$ (spatial context aggregation)
- $U_avg = (1/P^2)\sum U$ [pool over spatial axes]
- $\alpha = \sigma(\text{Conv1D}_k(U_avg))$ [depthwise 1D, k=3]
- $v = BN(\alpha \odot a + (1-\alpha) \odot b)$
-
- $\alpha + (1-\alpha) \equiv 1$ → ZERO redundancy between branches
- Only 80 parameters (0.03%) for the gating layer

Key Insight: Competitive Resource Allocation

- When $\alpha \to 1$: spectral fidelity dominates (suppress spatial blur)
- When $\alpha \to 0$: spatial context dominates (homogeneous regions)
- Forces each branch to learn non-overlapping representations
- Unlike A2S2K-ResNet / CSCANet that share a single gate

Followed by Spectral Deepening

- Fused tensor v enters 2 cascaded 3D Spectral ResBlocks
- Each: 1×1×7 Conv3D + ECA-3D efficient channel attention
- Residual skip connections preserve gradient flow
- 7,364 parameters (2.39%) for spectral refinement

Module 3: Second-Order Covariance Fusion Gate (SCFG)

Covariance-Based Channel Calibration

- Transitions from spectral → spatial processing stage
- Adaptive 3D pool collapses spectral dim: $C_2 \rightarrow 1$
- 1×1×1 Conv3D expands channels: K=16 → C_mid=128
- Squeeze singleton: spatial feature map x ∈ ℝ^(B×128×P×P)
-
- Second-Order Statistics:
- f_ch(b,c) = (1/N)∑ x(b,c,n) [channel descriptor]
- f_sp(b,n) = (1/C)∑ x(b,c,n) [spatial descriptor]
- G = f_ch ⊗ f_sp ∈ ℝ^(B×C×N) [outer product]
-
- $\tilde{G} = G / (\|G\|_F + \varepsilon)$ [Frobenius norm]
- g = σ(W_G · $\tilde{G}$) ∈ ℝ^(B×C_mid) [linear projection]
- x_scfg = x ⊙ g [spatial modulation]

Why Second-Order > First-Order

- Standard GAP computes μ = (1/N)∑x → loses co-activation patterns
- Outer product captures bilinear channel×spatial correlations
- Edges vs. interior textures have different co-occurrence patterns
- No iterative matrix square-root (unlike bilinear pooling nets)

Parameter Efficiency

- SCFG requires only 82 parameters (0.03% of network)
- Single learnable projection W_G ∈ ℝ^(1×N)
- Replaces dense FC layers → 85% parameter reduction
- Followed by 2 spatial ResBlocks (249,562 params, 81%)

Minority-Aware Hybrid Loss Function

Spread Loss (weight = 0.7)

- Penalizes when non-target activations intrude on true-class margin
- Dynamic margin $m(r) = 0.2 + 0.7 \cdot r$, scheduled linearly over training
- Early epochs: relaxed margin (m=0.2) prevents over-penalization
- Late epochs: strict margin (m=0.9) enforces wide class boundaries

Focal Loss (weight = 0.3)

- Inverse-frequency class weights: $w_c = 1/(N_c + \varepsilon)$
- Focusing parameter γ=2.0 down-weights easy-to-classify samples
- Prevents majority classes (Soybean-mintill: 2,468 samples)
- from dominating gradient updates over Oats (20 samples)

Combined Objective: $\mathscr{L} = 0.7 \cdot \mathscr{L}_{Spread} + 0.3 \cdot \mathscr{L}_{Focal}$

- Spread Loss enforces wide inter-class margins → prevents boundary collapse on minority classes
- Focal Loss re-weights gradient contributions → minority classes receive proportionally stronger updates
- Result: SSAN collapses to 64.19% OA (0% on Alfalfa/Oats); SGSCF-Net preserves 95.00% Alfalfa, 54.17% Oats
- Ablation: Switching to standard Cross-Entropy drops OA from 98.05% → 97.26% and AA from 94.82% → 93.78%

Experimental Setup & Benchmark Datasets

Indian Pines

AVIRIS sensor, 145×145 px
200 bands, 16 classes
10,249 labeled pixels
Severe imbalance (Oats: 20)

Pavia University

ROSIS-03 sensor, 610×340 px
103 bands, 9 classes
42,776 labeled pixels
Urban material confusion

Salinas Valley

AVIRIS sensor, 512×217 px
204 bands, 16 classes
54,129 labeled pixels
Fine-grained crop stages

Standardized Evaluation Protocol

- Split: 10% Train / 10% Validation / 80% Test (stratified per class)
- Patch size: 11×11 spatial neighborhood
- Batch size: 32, Epochs: 200
- Optimizer: Adam, LR: $5\times10^{-4} \rightarrow 10^{-6}$ (Cosine Annealing)
- Hardware: NVIDIA Tesla T4 GPU, 16 GB VRAM (Kaggle)

8 Baseline Models (Official Implementations)

- 3D CNNs: SSRN, A2S2K-ResNet
- Transformers: ViT, HSI-SSFTT, HyperRoPE-HiT
- Attention Networks: SSAN
- Capsule / Hybrid: Deep Matrix Capsules (DMC), CRV-MHDL
- All trained from scratch under identical 10/10/80 protocol

Results: Indian Pines — Per-Model Comparison

Model	OA (%)	AA (%)	κ	Time (min)
SSRN	**98.52**	79.99	0.9831	6.29
A2S2K-ResNet	98.11	94.41	0.9785	14.89
HyperRoPE-HiT	94.07	89.37	0.9324	1.43
HSI-SSFTT	98.45	94.29	0.9823	1.05
DMC	82.55	64.11	0.7989	17.73
ViT	90.33	89.12	0.8895	1.68
CRV-MHDL	67.20	51.71	0.6222	1.33
SSAN	64.19	72.77	0.5979	14.00
SGSCF-Net (Ours)	**98.05**	**94.82**	**0.9778**	**8.80**

Key Finding

- SSRN tops OA (98.52%) but scores
- 0% on Alfalfa, Grass-pasture-mowed, Oats
- SGSCF-Net: BEST AA (94.82%)
- across all 9 models evaluated

Minority Class Accuracy

- Alfalfa (46 px): 95.00%
- Grass-pasture-mowed (28 px): 95.24%
- Oats (20 px): 54.17%
- vs. SSRN: 0%, 0%, 0% on these
- vs. SSAN: 10.81%, 9.09%, 18.75%

Results: Pavia University — Urban Material Classification

99.82%	99.71%	99.76	12.9 min
Overall Accuracy	Average Accuracy	Kappa (κ × 100)	Training Time

Per-Class Highlights

- 4 classes at 100.00%: Asphalt, Painted Metal, Bare Soil, Bitumen
- Meadows: 100.00% (14,919 test pixels classified perfectly)
- Gravel: 99.76% — historically confused with Asphalt
- Self-Blocking Bricks: 99.32%
- Shadows (947 px): 99.34% via SCFG covariance operator
-
- CBSA α-gating isolates spectral absorption signatures:
- 1×1×7 branch separates Asphalt vs Bitumen hydrocarbon bands
- 3×3×7 branch preserves building structural boundaries

Baseline Comparison

- SSRN: 99.87% OA (marginally higher, but slower at 11.36 min)
- A2S2K-ResNet: 99.83% OA, 41.12 min training (3.2× slower)
- DMC: 99.34% OA, 73.42 min (5.7× slower, EM routing)
- SSAN: 76.82% OA — severe collapse on Gravel (17.96%)
- CRV-MHDL: 98.40% OA — degrades on boundary classes
-
- SGSCF-Net: competitive accuracy at 12.9 min
- with only ~308K params (vs. ~380K A2S2K, ~6.4M SSAN)

Results: Salinas Valley — Fine-Grained Agriculture

99.92%	99.92%	99.91	57.7 min
Overall Accuracy	Average Accuracy	Kappa (κ × 100)	Training Time

Per-Class Highlights

- 11 of 16 classes reach 100.00% accuracy
- All 4 Romaine lettuce growth stages (4-7 wk): 99.65-100%
- Vineyard untrained: 99.73% vs. Vineyard trellis: 100.00%
- SCFG captures chlorophyll absorption band-ratio shifts
-
- Fine-Grained Discrimination:
- Lettuce cultivars vary only in narrow chlorophyll depth
- CBSA dual-branch isolates spectral vs. canopy geometry
- Bilinear covariance amplifies subtle band-ratio differences

Baseline Comparison

- A2S2K-ResNet: 100.00% OA but 10.34 min (similar speed)
- SSRN: 99.96% OA, 30.98 min (slower)
- SSAN: 88.20% OA — Grapes untrained 57.89%, collapse
- CRV-MHDL: 96.30% — Vineyard untrained confusion
-
- Near-perfect performance confirms generalizability
- across diverse spectral-spatial complexity levels
-
- OA rises: 98.05% (IP) → 99.82% (PU) → 99.92% (SV)
- as minimum per-class sample count grows

Visual Comparison: Classification Maps

[Classification maps from Figures 3, 4, 5 — Ground Truth vs. all 9 models across 3 datasets]

[Insert Fig. 3 — Indian Pines Maps]

[Insert Fig. 4 — Pavia University Maps]

[Insert Fig. 5 — Salinas Valley Maps]

Visual Analysis Summary

- SGSCF-Net produces the most spatially homogeneous classification maps across all 3 datasets
- SSAN shows pervasive salt-and-pepper noise; CRV-MHDL and DMC erase minority parcels entirely; SSRN blanks out Alfalfa/Oats
- SBAM filtering + CBSA complementary gating yields crisp parcel boundaries with suppressed atmospheric scatter

Computational Efficiency & Parameter Analysis

Model	Params	Checkpoint	IP Time	OA (%)
ViT	~610K	2.4 MB	1.68 min	90.33
HSI-SSFTT	~156K	625 KB	1.05 min	98.45
A2S2K-ResNet	~380K	1.5 MB	14.89 min	98.11
SSAN	6.40M	—	14.00 min	64.19
DMC	~480K	—	17.73 min	82.55
SGSCF-Net (Ours)	**308,097**	**1.2 MB**	**8.80 min**	**98.05**

Test Inference Speed

- Indian Pines: 1.18 seconds
- Pavia University: 4.12 seconds
- Salinas Valley: 5.46 seconds
-
- Order of magnitude faster than:
- DMC: 62.70s on Salinas (11.5× slower)
- SSAN: 135.74s on Salinas (24.9× slower)
-
- Linear O(N) complexity vs $O(N^2)$ attention

Module-Level Parameter Breakdown (Indian Pines)

- SBAM: 10,225 (3.32%) | CBSA: 80 (0.03%) | SCFG: 82 (0.03%) → 3 novel modules = only 10,387 params (3.37% of total)
- Primary volume: Spatial 3D entry (11.98%) + Spatial ResBlocks (81.00%) — standard Conv3D backbone
- Classifier head: 272 params (0.09%) — single linear layer, no deep FC stacks

Ablation Study & Block Depth Sensitivity

#	SBAM	CBSA	SCFG	H.Loss	OA (%)	AA (%)
1	✓	×	×	×	93.65	89.24
2	✓	✓	×	×	95.18	91.80
3	✓	✓	✓	×	96.44	93.05
4	✓	✓	✓	CE	97.26	93.78
5	**✓**	**✓**	**✓**	**✓**	**98.05**	**94.82**

Block Depth Sensitivity (D)

- D=1: 179K params, 95.82% OA (underfits)
- D=2: 308K params, 98.05% OA (optimal)
- D=3: 437K params, 97.88% OA (diminishing returns)
- D=4: 566K params, 97.40% OA (overfitting onset)
-
- D=2 yields best accuracy-efficiency trade-off

Ablation Insights

- SBAM alone lifts baseline from ~90% to 93.65% by filtering atmospheric noise before convolution
- Replacing shared SE gate with complementary CBSA adds +1.53% OA by forcing orthogonal branch specialization
- SCFG contributes +1.26% by recovering second-order spatial-channel correlations discarded by first-order GAP
- Hybrid loss adds final +0.79% OA and +1.04% AA — critical for minority-class boundary protection (Oats: 0% → 54.17%)

60-Run Hyperparameter Grid Search

5 Patch Sizes (9×9, 11×11, 13×13, 15×15, 17×17) × 4 Batch Sizes (32, 64, 128, 256) = 20 runs/dataset × 3 datasets = 60 total

[Fig. 7 Heatmap — Indian Pines]

[Fig. 7 Heatmap — Pavia University]

[Fig. 7 Heatmap — Salinas Valley]

Optimal Configuration

- Patch 11×11, Batch 32
- OA: 98.05% (IP), 99.82% (PU), 99.92% (SV)
- Consistent Rank 1 or within 0.08%
- Training: 8.80 min (IP)

Spatial Window Trade-off

- 9×9: underfits (misses texture boundaries)
- 15×15, 17×17: overfits (cross-field contamination)
- 11×11 at 20m/pixel = 220m ground distance
- Captures field-level context without noise

Batch Size Dynamics

- B=32 beats B=256 by 3.43% on Indian Pines
- Small batches: higher stochastic gradient variance
- Helps optimizer escape sharp local minima
- Preserves minority class decision boundaries

Data Scarcity Stress Testing: 1% – 10% Training Ratios

Dataset	1% Train	2% Train	5% Train	10% Train	Drop
Indian Pines	70.12%	81.55%	93.91%	97.84%	-27.73%
Pavia University	95.41%	98.04%	99.51%	99.83%	-4.43%
Salinas Valley	97.73%	99.11%	99.79%	99.93%	-2.20%

Salinas: Most Resilient (-2.20%)

- 97.73% OA with just 1% training data
- Large contiguous fields enable SBAM/SCFG
- to extract representative descriptors
- from minimal training patches

Pavia: Moderate Drop (-4.43%)

- 95.41% OA at 1% despite urban material
- ambiguity (Asphalt vs Bitumen confusion)
- CBSA gating preserves distinct spectral
- signatures even with sparse supervision

Indian Pines: Steepest Drop (-27.73%)

- 70.12% OA at 1% — Oats has only 1-2 patches
- Mathematical floor: can't learn 3D features
- from < 2 spatial patches per class
- Not an architectural defect but a data limit
- Hybrid loss still prevents total collapse

Cross-Dataset Summary & Key Achievements

98.05%	99.82%	99.92%	308K
Indian Pines OA	Pavia Univ. OA	Salinas Valley OA	Total Parameters

Project Achievements

- Designed & implemented 3 novel modules (SBAM, CBSA, SCFG)
- Best Average Accuracy on Indian Pines (94.82%) across all 9 models
- 99.82% OA on Pavia, 99.92% OA on Salinas (near-perfect)
- 120+ GPU runs: 60 grid search + 12 scarcity + 24 baseline + proposed
- O(N) linear complexity — no quadratic self-attention overhead
- 1.18s test inference on Indian Pines (vs. 135.74s SSAN)
- Complete 31-page ACM-format research manuscript drafted

Why SGSCF-Net Matters

- Lightweight: 308K params vs. 6.4M SSAN, ~610K ViT, ~480K DMC
- Balanced: Protects minority classes where SSRN/SSAN collapse to 0%
- Generalizable: Consistent top-tier across 3 diverse HSI scenes
- Efficient: 8.80 min training vs. 17.73 (DMC), 41.12 (A2S2K)
- Modular: SBAM/CBSA/SCFG are plug-and-play for any 3D CNN
- Reproducible: Fixed 10/10/80 split, 200 epochs, public datasets
- Edge-deployable: 1.2 MB checkpoint, suitable for UAV/satellite

Future Work & Research Directions

1 Hardware-Aware Quantization

INT8/FP8 tensor quantization + structured pruning for real-time inference on edge NPUs and FPGAs aboard UAVs and satellite platforms

2 Self-Supervised Pretraining

Masked spectral pre-training on large uncurated HSI cubes to learn transferable representations and reduce label dependency below 1%

3 Multi-Modal Fusion

Extend SCFG covariance mechanism to joint HSI + LiDAR classification for multi-sensor Earth observation pipelines

4 Journal Publication

Target IEEE Transactions on Geoscience and Remote Sensing (TGRS) or similar Q1 journal for peer-reviewed publication of this work

Thank You

Questions & Discussion

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National Institute of Technology, Raipur • Department of Information Technology • November 2026